

Mathematical Methods for International Commerce

Week 14/1: Cramer's Rule (7.3)

Igor Vyshnevskyi, Ph.D.

Sogang University, Geppert School of International Studies

June 2, 2026

Agenda

1. Cramer's Rule (7.3)
2. Group activity: Solving matrixes
3. Homework #2



The seal of Gangneung University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. Above the shield is the year '1960'. The outer ring of the seal contains the university's name in Korean and English.

1. Cramer's Rule (7.3)

Why It Matters in Economics, Business & Finance

- Many real-world economic systems are governed by **simultaneous equations**.
- Think of equilibrium conditions in macroeconomic models, or supply-demand analysis in trade.
- **Cramer's Rule** is a neat algebraic technique for solving linear systems using **determinants**, without explicitly computing the inverse matrix.

You'll learn to:

- See when and why Cramer's Rule is useful
- Use it to solve 2×2 and 3×3 systems
- Apply it to real examples in **macroeconomics** and **international trade**

Recap: Linear System as Matrix Equation

Given a system:

$$a_{11}x + a_{12}y = b_1$$

$$a_{21}x + a_{22}y = b_2$$

It can be written as:

$$Ax = b, \quad \text{where} \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad x = \begin{bmatrix} x \\ y \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Usually, we solve this using the inverse:

$$x = A^{-1}b$$

But when A^{-1} is hard to compute, **Cramer's Rule** offers a shortcut.

Cramer's Rule: The Concept

Let A be a square matrix with non-zero determinant. Then the solution x_i to the system $Ax = b$ is:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Where:

- A_i : the matrix formed by replacing the *i-th column* of A with the vector b
- $\det(A) \neq 0$: system has a unique solution
- Cramer's Rule works only for square systems (same number of equations and unknowns)

Example 1: Solve Using Cramer's Rule

Given:

$$4x + 3y = 10$$

$$2x + y = 4$$

Step 1: Coefficient matrix:

$$A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 10 \\ 4 \end{bmatrix}$$

Step 2: Determinant of A :

$$\det(A) = (4)(1) - (3)(2) = 4 - 6 = -2$$

Example 1: Solve Using Cramer's Rule (cont)

Step 3: Determinants with substituted columns:

$$A_1 = \begin{bmatrix} 10 & 3 \\ 4 & 1 \end{bmatrix}, \quad \det(A_1) = 10 \cdot 1 - 3 \cdot 4 = 10 - 12 = -2$$

$$A_2 = \begin{bmatrix} 4 & 10 \\ 2 & 4 \end{bmatrix}, \quad \det(A_2) = 4 \cdot 4 - 10 \cdot 2 = 16 - 20 = -4$$

Final Solution:

$$x = \frac{-2}{-2} = 1, \quad y = \frac{-4}{-2} = 2$$

Example 1: Solve Using Cramer's Rule (cont)

Quick Check

Verify the solution:

$$x = 1, \quad y = 2$$

Substitute into the original system:

$$4(1) + 3(2) = 10$$

$$2(1) + 2 = 4$$

Both equations hold ✓

Example 2: Solve Using Cramer's Rule

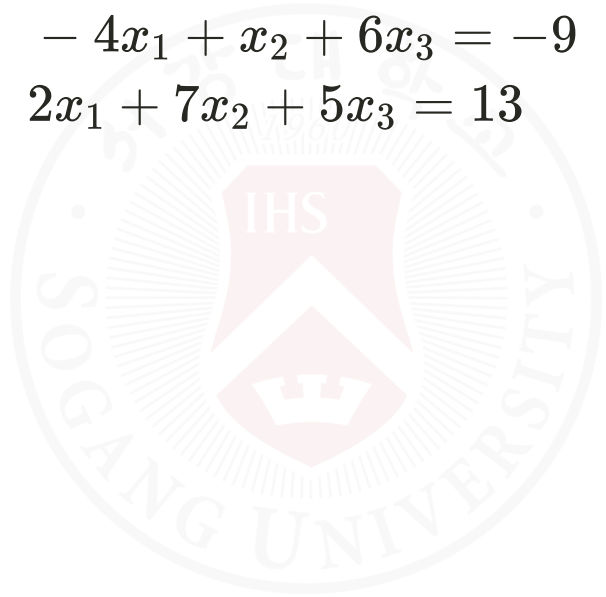
Solve the system:

$$x_1 + 2x_2 + 3x_3 = 9$$

$$-4x_1 + x_2 + 6x_3 = -9$$

$$2x_1 + 7x_2 + 5x_3 = 13$$

Find x_1 using **Cramer's Rule**.



Step 1: Identify Matrices

Coefficient matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ -4 & 1 & 6 \\ 2 & 7 & 5 \end{bmatrix}$$

Right-hand side vector b :

$$b = \begin{bmatrix} 9 \\ -9 \\ 13 \end{bmatrix}$$

Replace the **first column** of A with b to form A_1 :

$$A_1 = \begin{bmatrix} 9 & 2 & 3 \\ -9 & 1 & 6 \\ 13 & 7 & 5 \end{bmatrix}$$

Step 2: Compute Determinants

Expand $\det(A_1)$ along the top row:

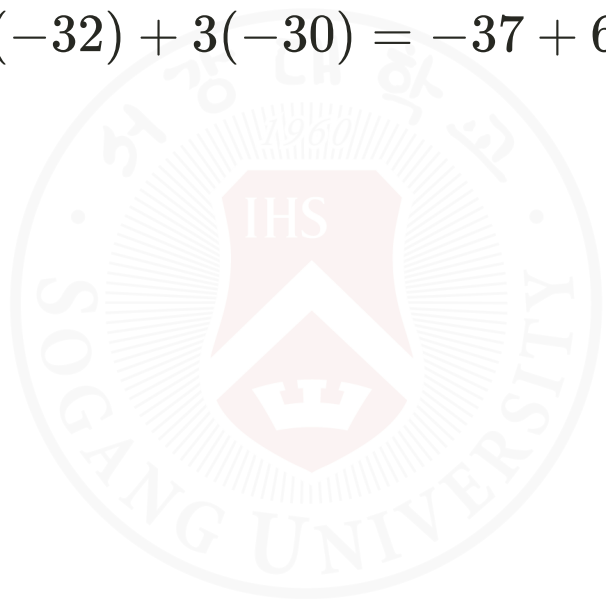
$$\begin{aligned}\det(A_1) &= 9 \cdot \begin{vmatrix} 1 & 6 \\ 7 & 5 \end{vmatrix} - 2 \cdot \begin{vmatrix} -9 & 6 \\ 13 & 5 \end{vmatrix} + 3 \cdot \begin{vmatrix} -9 & 1 \\ 13 & 7 \end{vmatrix} \\ &= 9(-37) - 2(-123) + 3(-76) = -333 + 246 - 228 = -315\end{aligned}$$

Recall:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Step 3: Compute $\det(A)$

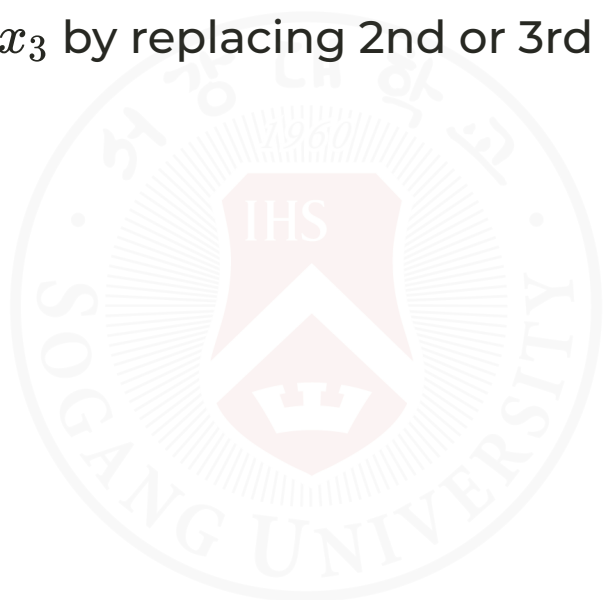
$$\begin{aligned}\det(A) &= 1 \cdot \begin{vmatrix} 1 & 6 \\ 7 & 5 \end{vmatrix} - 2 \cdot \begin{vmatrix} -4 & 6 \\ 2 & 5 \end{vmatrix} + 3 \cdot \begin{vmatrix} -4 & 1 \\ 2 & 7 \end{vmatrix} \\ &= 1(-37) - 2(-32) + 3(-30) = -37 + 64 - 90 = -63\end{aligned}$$



Step 4: Apply Cramer's Rule

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{-315}{-63} = 5$$

- Use same method to find x_2, x_3 by replacing 2nd or 3rd column of A .



Real-World Example: Static Macroeconomic Model

Consider a simple Keynesian model:

$$Y = C + I + G$$
$$C = 100 + 0.8Y$$

Solve for Y and C when $I = 50$ and $G = 20$

Step 1: Substitute:

$$Y = (100 + 0.8Y) + 50 + 20 \Rightarrow Y = 170 + 0.8Y$$

$$Y - 0.8Y = 170 \Rightarrow 0.2Y = 170 \Rightarrow Y = 850$$

$$C = 100 + 0.8 \cdot 850 = 780$$

Rearranging:

$$Y - C = 70$$

$$-0.8Y + C = 100$$

Expressing this in matrix form:

$$A = \begin{bmatrix} 1 & -1 \\ -0.8 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} Y \\ C \end{bmatrix}, \quad b = \begin{bmatrix} 70 \\ 100 \end{bmatrix}$$

Cramer's Rule Application

Apply Cramer's Rule to confirm.



Example 2: Two-Country Trade Model

Country A and B produce goods X and Y.

- $2X + 3Y = 20$ (A's resource constraint)
- $3X + Y = 18$ (B's resource constraint)

Solve for quantities of X and Y that satisfy both.

Matrix Form:

$$A = \begin{bmatrix} 2 & 3 \\ 3 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 20 \\ 18 \end{bmatrix}$$

Compute determinants:

$$\det(A) = 2 \cdot 1 - 3 \cdot 3 = 2 - 9 = -7$$

$$A_1 = \begin{bmatrix} 20 & 3 \\ 18 & 1 \end{bmatrix}, \quad \det(A_1) = 20 \cdot 1 - 3 \cdot 18 = 20 - 54 = -34$$

Example 2: Two-Country Trade Model (cont)

$$A_2 = \begin{bmatrix} 2 & 20 \\ 3 & 18 \end{bmatrix}, \quad \det(A_2) = 2 \cdot 18 - 20 \cdot 3 = 36 - 60 = -24$$

Solution:

$$X = \frac{-34}{-7} \approx 4.86, \quad Y = \frac{-24}{-7} \approx 3.43$$

Practice Problems: Solve using Cramer's Rule

- Problem 1:

$$3x + 4y = 10$$

$$2x + y = 5$$

- Problem 2:

$$2x + 3y - z = 5$$

$$4x - y + 2z = 6$$

$$-3x + 2y + z = -4$$

- Problem 3: In a 3-sector macro model:

$$Y = C + I + G$$

$$C = a + bY$$

Set $a = 120$, $b = 0.75$, $I = 50$, $G = 30$. Solve for Y and C .

- Problem 4: Trade model:

$$4X + 5Y = 40$$

$$6X + 2Y = 38$$

Use Cramer's Rule to find X , Y .

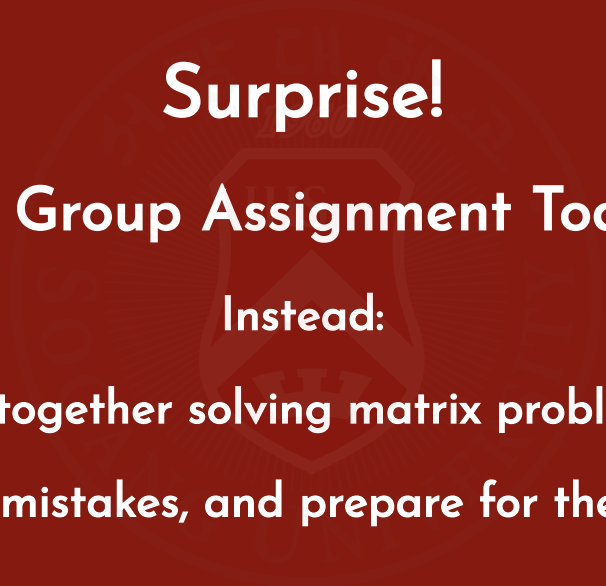
Summary

- Cramer's Rule offers a quick solution for **small** systems using determinants
- Works only if $\det(A) \neq 0$
- Particularly useful for **economic models** involving linear constraints
- Cramer's Rule becomes inefficient for large systems because determinant calculations become complicated

Limitations: Only practical for small systems due to cost of computing determinants.

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2. Group activity: Solving matrices



Surprise!

No Group Assignment Today

Instead:

We will practice together solving matrix problems step-by-step

Ask questions, make mistakes, and prepare for the quiz/exam calmly :)

3. Home work #2



Homework #2

- **Due Date:** June 11, 2026, before the start of class.
- **Submission Format:** Submit your solutions as **a single file** via the Cyber Campus.
- **Instructions:**
 - Clearly show all steps and calculations.
 - Include explanations for your answers where applicable.
 - Ensure your submission is neat and well-organized.
 - Bring any questions to the office hours or email me.
- Work on your Home Assignment #2 (Jacques, 10th edition, Chapters 5-9):
 - Chapter 5.1: Exercise 5.1, Problem 5 (p. 411)
 - Chapter 5.2: Exercise 5.2, Problem 7 (p. 427)
 - Chapter 5.3: Exercise 5.3, Practice Problem inside the chapter (p. 432 only)
 - Chapter 5.4: Exercise 5.4, Problem 5 (p. 456)
 - Chapter 5.5: Exercise 5.5, Problems 6 and 7 (p. 467)
 - Chapter 5.6: Exercise 5.6, Problem 4 (p. 479)
 - Chapter 6.1: Exercise 6.1, Problem 3 (p. 506)
 - Chapter 6.2: Exercise 6.2, Problem 6 (p. 521)
 - Chapter 7.1: Exercise 7.1, Problem 5 (p. 553)
 - Chapter 7.2: Exercise 7.2, Problem 6 (p. 572)
 - Chapter 7.3: Exercise 7.3, Problem 5 (p. 584)
 - Chapter 8.1: Exercise 8.1, Problem 6 (p. 615)
 - Chapter 8.2: Exercise 8.2, Problem 3 (p. 626)
 - Chapter 9.1: Exercise 9.1, Problem 4 (p. 655)
 - Chapter 9.2: Exercise 9.2, Problem 3 (p. 670)

Good luck!

Next Class

- (June 4) Linear Programming (8.1)





Any QUESTIONS?

Thank you for your attention and participation!